Process Report – StudyHelper

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TABLE OF CONTENTS

1. INTRODUCTION

Provide a factual, chronological overview of the project from kick-off to submission, grounded in concrete sources (logbook entries, meeting minutes, sprint retrospectives). Highlight major milestones and any significant pivots in direction.]

We chose the project StudyHelper probably because of the relevance of the topic in all our lives. The idea of a device that can help navigate in choosing the best environment for studying was at the intersection of our interests and feasibility within the project scope.

The initial phase of the project involved heavy brainstorming sessions and discussions. The main point of these debates was refining our problem statement to ensure it would be meaningful to solve with a solution that would be technically relevant for all sub-teams in a way that satisfies the overall expectations of the project.

2. GROUP WORK

2.1 Group Composition and Profiles

[Introduce each group member briefly — background, relevant skills, and prior experience. If the group is multicultural or interdisciplinary, highlight this and reflect on how it influenced the collaboration.]

Member	Background	Key Strengths	Role(s) in Project
[Name]	[Study/origin]	[Skills]	[e.g. Backend, PM]

2.2 Roles and Contributions

[Describe clearly how responsibilities were divided. Who led which areas? Was the division planned or did it emerge organically? Provide specific examples: “X took ownership of the database layer and led the sprint planning sessions in weeks 3–6.”]

2.3 Team Dynamics and Collaboration

[How did the team communicate day-to-day? What tools did you use (Discord, Jira, etc.)? Were there informal leadership dynamics? How were decisions made — by consensus, by vote, or by domain ownership?]

2.4 Conflict and Resolution

[Did any disagreements or tensions arise? How were they addressed? What did the team learn from handling conflict? If no significant conflict occurred, briefly note how the group maintained alignment.]

2.5 Social Loafing and Accountability

[Were there any instances where workload felt unevenly distributed? How did the group ensure accountability? What mechanisms (standups, task boards, peer review) helped keep everyone engaged?]

3. PROJECT INITIATION

3.1 Choosing the Problem Domain

[Why did your group choose this particular topic? Was it driven by personal interest, practical relevance, available resources, or suggestions from a supervisor/company? In hindsight, was this a good choice? What would you do differently?]

3.2 Problem Statement Development

[How did you arrive at the final problem statement? Describe the iterations. Was the initial framing too broad, too narrow, or off-target? What clarified your thinking — literature, supervisor input, prototyping?]

3.3 Time Planning and Scheduling

[How did you plan the project timeline? Did you use sprints, Gantt charts, or a looser milestone structure? How realistic was the initial plan? Reflect on deviations: what caused delays, and how were they managed?]

3.4 Ethical Perspectives

[What ethical questions arise from the problem your project addresses? Consider: data privacy, environmental impact, fairness, accessibility, or societal effects. Were these considerations built into the project from the start, or addressed reactively?]

4. PROJECT EXECUTION

This section describes the development process, divided into our collective efforts and team-specific contributions.

4.1 Together Process

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[Describe how the whole group worked together during the execution phase. How were the components (IoT, ML, Frontend) integrated? How did we handle cross-team dependencies and API contracts? Reflect on the effectiveness of our shared git workflow and communication.]

4.2 IOT Team Process

[... Describe the development of the embedded C code, hardware assembly, and testing. ...]

4.3 MAL Team Process

The Machine Learning and API (MAL) development followed an iterative path from data exploration to pipeline implementation. The following notes capture the key decisions and observations made during this process.

4.3.1 Mock Data Search and Goal Refinement

Initially, the focus was on how environmental noise influences focus. We attempted to combine datasets of focus-related noise effects with labeled sound categories from WAV files, extracting frequencies and loudness. However, we realized that “focus” cannot be measured directly.

Consequently, we narrowed the goal to predicting a user-provided **Study Suitability Rating**, which made the target variable more grounded in user feedback.

4.3.2 Data Quality and Correlation Analysis

Further investigation led to the elimination of several datasets that appeared synthetic. We observed that in some candidate datasets, the distributions of features like humidity, noise, and light were suspiciously uniform, suggesting algorithmic generation rather than real sensor collection.

We also analyzed feature correlations as part of our validation process. Healthy datasets showed natural physical correlations (e.g., CO_2, temperature, and humidity), while suspicious datasets exhibited either zero correlation or extreme overfitting potential. We decided to merge diverse datasets to create a more robust mock dataset.

4.3.3 Advanced Imputation Strategy

To handle missing values after merging, we implemented a sophisticated approach using the **MICE (Multivariate Imputation by Chained Equations)** framework. We enhanced this by:

- **Clustering**: Using k-means to find environment types (e.g., sessions made with high sun exposure vs. sessions made in labs).
- **ExtraTrees Estimator**: Modeling complex non-linear relationships.
- **Variance Modification**: Including natural distribution variance to avoid "flat average" imputation.
- **Global Median Fallback**: Used for extreme sparsity to prevent bias.

4.4 Frontend Team Process

[... Describe the development of the React application, UI/UX iterations, and integration with the backend API. ...]

4.5 Application of Methods and Theories

[Reflect on the engineering and analytical methods you applied (requirements analysis, architecture design, testing frameworks, etc.). Were the methods well-suited to the problem? Did you feel confident using them, or were there learning curves? What worked well and what fell short?]

4.6 Challenges During Execution

[Describe the most significant technical or organisational challenges encountered. How did the team respond? What was the outcome? Examples: “The I2C sensor driver took much longer than expected,” or “The API contract changes caused significant rework in the frontend.”]

4.7 Deviations from the Plan

[Compare what was planned versus what was actually executed. What changed and why? Were changes proactive (intentional pivots) or reactive (forced by circumstances)? How did the team adapt?]

4.8 Use of Tools and Technologies

[Reflect on the tools and technologies used throughout the project (IDEs, version control, project management, testing frameworks, etc.). Were they effective? Were there tools you wished you had used, or tools you regret choosing?]

5. PERSONAL REFLECTIONS

6.1 The Supervision Process

[Describe how supervision was conducted. How often did you meet? How did you prepare for meetings? What format did sessions typically take?]

6.2 Impact on the Project

[In what specific ways did supervision change or improve the project? Were there suggestions or challenges from the supervisor that redirected your work? Were there times you disagreed with feedback — how did you handle that?]

6.3 Lessons for Future Projects

[What will you do differently in your next supervised project? Examples: prepare more structured agendas, present partial results earlier, ask more targeted questions, be more proactive about seeking feedback. Be concrete and actionable.]

7. CONCLUSION

7.1 Group Summary

[Summarise the group's overall experience. What defined this project's process? What are the most important things you collectively learned — about software engineering practice, teamwork, or the PBL learning model? How has this project prepared you for professional work?]

7.2 Recommendations

[Produce a practical list of do's and don'ts for future groups undertaking a similar project. These should be grounded in your actual experience — not generic advice.

Do:

- ...
- ...

Avoid:

- ...
- ...]

8. REFERENCES

- Vroom, V. H. (1964). Work and motivation .
- Darley, J. M., & Latane, B. (1968). Bystander intervention in emergencies: Diffusion of responsibility. Journal of Personality and Social Psychology , 8 (4, Pt.1), 377–383. **https://doi.org/10.1037/h0025589**
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